

Time Will Tell: The Case for an Idiographic Approach to Behavioral Cybersecurity Research

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Mayukh Mukhopadhyay: *Could you share any unexpected challenges or interesting insights you encountered while conducting the empirical study for this research article?*

Alexander Benlian: I think one of the key challenges in our journey to publish this article was that our starting point was aimed at writing a methods paper regarding the idiographic approach.

If I remember correctly, on a broader level, we initially focused on the topic of cybersecurity fatigue, which we wanted to use as a sample study to showcase the idiographic approach.

During the review process over the rounds of revisions, based on the feedback, a key challenge was that we would be better off transforming this methods article into a research article and focusing, on a broader and more established cybersecurity theory to showcase the idiographic approach.

So, the challenge was maybe we started out with quite a current topic of cybersecurity fatigue and then, came up with a more classical theoretical approach in the cybersecurity literature. And this was suggested maybe because the reviewers and the editors thought that it would make a better example for the benefits of the idiographic approach if we are able to show that we can validate part of existing between-person theories, but also extend, the classical approaches. That is why we shifted to the neutralization study, over time. That was a bit of an unexpected challenge, to shift so much during one review process.

John D'Arcy: That is when Alex (Alexander Benlian) talks about the classic theories used in the behavioral cybersecurity research realm. Neutralization is well-known and we initially started out with this idea of looking at security fatigue since it was kind of a new concept.

As Alex said, I remember the reviewers said, why don't you pick a theory that is more well-known in this space to sort of prove your point about the benefits of the idiographic approach?

This is sort of a lesson learned even for authors who might be listening to this as Alex said, we started out with this being a methods article, and it completely changed in terms of our approach.

We can talk a little bit more about that, but we were also challenged to add a little bit more of a theoretical contribution as, or initially we looked at it as just sort of a methods piece and saying. We think this idiographic approach is interesting and something that maybe not, hasn't been highlighted too much in the behavioral cybersecurity realm. So we really shifted and we had a great, review experience. They (the reviewers) were pretty clear and direct and from what I recall, as with most papers. I think we can say that it definitely ended up better as a result of the process.

Interviewers:

Thank you, Professor.

Mayukh: *What inspired you to collaborate on this research project, and what common thread sustained your teamwork throughout the writing process?*

Since writing single-handedly is something different and collaborating together wherein three authors from three different universities, how, what inspired, and what was the common threading?

John: This is really where I think the benefits of going to conferences and getting to know people really came to fruition because W. Alec Cram (first author of this paper) and I had been working on some projects in the cybersecurity realm.

And I remember when we initially were thinking about this project, I thought we really could use someone with some expertise and sort of this within the individual approach, an idiographic approach. I think I said to Alec, I said, you had been collaborating with Alexander Benlian before on some project and I was aware that, Alex had done some research using experience sampling approaches. And I think just Alec and I, during one of our conversations, I said, maybe we should ask him if he wants to get involved and I really think that it would be a great addition to the team.

And so that's sort of how I remember it coming about in terms of the three of us coming together and working on this project.

Alexander: Yeah, maybe just add that a common thread was our curiosity in terms of how cybersecurity behavior changes over time.

I think John (John D'Arcy) also did a couple of early studies on within-person behaviors and I did also a couple of studies, not in the cybersecurity field, but in related research streams in the IS (Information Systems) field.

But I think a common thread was to find out more about how cybersecurity behaviors change within-person over time, and this was the key inspiration, I would say. Because we saw an opportunity to go down this path and shed more light on the fluctuations that may happen within persons over time in terms of security behavior.

Mayukh Mukhopadhyay: *Were there any particular findings or results from your study that surprised you or challenged your initial assumptions, leading to further exploration or insights on behavioral cybersecurity phenomena?*

John: The empirical study we have in there, is on neutralization theory and basically how people respond to it during different days of the week in terms of do they tend to neutralize more or less as the week goes on, as the work week goes on. I think we initially didn't have that.

We initially had our empirical study, we additionally had one about security fatigue and it was a longer time period. So we switched to this study, we were looking at just during the course of the week. And at least for me, I found it sort of surprising that we were finding that the theory, I guess the strength of the relationship between neutralization and employee security related behavior varied during the course of the week.

So we made that sort of one of our core contributions and what we describe as a contribution to our understanding of neutralization theory and how it applies to employee security behavior. But I would say initially it surprised me in a positive way and I thought this is really interesting. So maybe can we sort of theoretically explain why this might be happening?

So, that made it into the paper and I think it is one of the key contributions of the paper.

Alexander: Yeah, I have nothing to add. I think that is the key and surprising point that we found in our study. And that's also the core message of our paper that there can be novel findings that go above and beyond between-person findings.

Kaushik Ghosh: Thank you, professors. So I'll take over from here. So next questions are discourse and domain-specific.

In what ways does the idiographic approach, which involves within-person analysis of longitudinal data, contribute to the validation of widely accepted theories in behavioral cybersecurity research, which are primarily based on the nomothetic approach (i.e., cross-sectional data and between-person analysis)?

John: Do you want to take that, Alex?

Alexander: Yeah, I can take this one.

So in terms of validation, well, maybe it's worthwhile to just go back to the between-person approach, which you basically can, well, compare to getting snapshots across many individuals in one point in time.

So it's about a population of individuals. This is contrasted to the idiographic approach, the within-person approach, where it's not a snapshot that we want to get at, but more or less at some sort of movie. We are looking at individual behavior over time. So several measures over time, and to look at a population of experiences and situations that we are interested in, and we want to understand how stable and consistent cybersecurity behavior is over time within people.

And so in terms of validating, well, that's the question, what happens at the between-person level and what happens in the within-person level. And in a lot of cases, well, you can see also in other fields, not only in the IS field, that well, basic relationships on the between-person level can be, well, validated on the within-person level. That you have similar relationships in terms of direction, and significance. And well, this can be one goal, right, of idiographic approaches and studies that you try to find similar relationships. And I think it's the technical term is called homology, right, across levels.

And while this might be one goal, right, of your idiographic research, but of course, and well, we are also mentioning this in the paper, I think from a theoretical point of view, it's much more interesting to find diverging results on the different levels.

And this is what we, well, I think already mentioned in our study in terms of neutralization, that the relationship, right, that we found over the different work weeks got more, was reinforced, right, the strength increased over the work week as the work week progresses.

And this was something interesting that we see a shift in the significance of the effects where we have, on some days, no relationship, a null relationship, and on some other days, we have a significant relationship, right?

Well, if I remember correctly, a negative relationship.

John: Right, and that shifting relationship, we would not have been able to see if we did the nomothetic approach, the between-person approach, it would have been masked, it would just have been, all that would have been within there.

So that was the benefit of the idiographic approach, it allowed us, at least in that case, to dig in and see the differences in, how the theory works on different days of the week.

Interviewers: Yeah, thank you so much, professors.

Kaushik: *What are the specific variables and proxies employed to quantify the evolution of cybersecurity behavior among individuals (employees) over time?*

John: So what are specific, the question, what are the specific variables, how do you proxy cybersecurity behavior over time?

Kaushik: Yes.

John: I think for us, I'm trying, at least in terms of how we did it, it was more of a methodological approach.

Whereas, you know, we use some of the, if you look at our survey, we used in the study in the paper, we use basically the same survey measurement items, although the wordings were adjusted for daily measurement, and but it was more the experience sampling approach where we captured these measures at multiple points in time.

So making it a purely within-person study, and that allows us to explore this evolution of behavior over time.

So at least one way to answer your question, we didn't actually change the measures. It was more about how they were utilized, the methodology, and the research design. So that's one way to go about trying to get, you know, cybersecurity behavior over time, just capturing it at multiple points in time, as the experience sampling approach does.

If you have anything different to say on that, Alex?

Alexander: No, I totally agree.

I think we also show in Table five of our article that there are a couple of methods, analytical, but also data collection methods, in terms of how you can quantify the evolution of constructs, there are several alternatives. And we picked one, but there, for sure, there are a couple of other methods on how you can model trajectories or the evolution of phenomena and relationships over time. So there are a couple of alternatives.

Kaushik: *How does the severity of sanctions influence cybersecurity compliance within an organization? How is this effect measured in the current paper, and what additional measures could be implemented to enhance compliance?*

Alexander: I cannot remember whether we have measured this in our, in our, in this empirical study. I think in the neutralization study, we did not focus on the severity of sanctions. I think this is not, not included in this research model.

So John, correct me if I'm wrong, but I think the severity of sanctions is more integrated in deterrence theoretical model research models, right?

John: Yeah, and I've done some research in that area with deterrence models.

But to answer your question, I don't think we have it in this. When I initially saw the question, I was like, maybe we had it as a control variable. It's often considered as a control variable in studies where they look at employee compliance. But I don't think it made it into this one.

But you asked the question, how does the severity of sanctions influence cybersecurity compliance? There's a lot of research in the behavioral cybersecurity realm that uses deterrence theory. And the basic idea is if people perceive stronger sanctions for non-compliance, they are then more likely to engage in compliant behavior.

It's a pretty simple theoretical perspective. There are multiple ways with it. You can dig into the theory, deterrence theory. And when you look at perceptions of sanctions, you can look at the certainty of sanctions.

So how likely sanctions are likely to occur, the severity, so how severe they are as you asked about? Then there's also this concept of celerity, which is how swift the sanctions are. Typically those three dimensions of sanctions are what are looked at in these deterrence-based studies.

I say a lot, there's a pretty significant body of research, particularly some of the earlier stuff that's looked at employee security behavior in the IS literature that's based on this, these deterrence principles, and they do look at the influence of severity of sanctions.

So, yeah, it wasn't a key focus of this study. Like I said, I can't recall if we if we ended up using it as a control, because it sort of is a known predictor of employees compliant or non-compliant behavior. And that is, of course, the dependent variable that we were looking at.

I also don't think so, go ahead, Alex.

Alexander: Yeah, just one comment.

I think what would be interesting from an idiographic perspective would be, or what would be the more interesting question, how the severity of sanctions changes over time within organizations, and how these changes have, in turn, an effect on cybersecurity behaviors.

So I think this would, well, this will be an interesting question, research questions, going forward.

John: Yeah, right, we focus on neutralization theory, but I think probably you could think of maybe applying an ideographic approach to deterrence theory and seeing how if we can gain additional insights from that theory into that theory in this realm.

Kaushik: Yes, thank you, sir.

In one table, I could remember it was, I mean, five, six, and seven, such kind of numbers were given over which measures the behavior, which talks about the severity of sanctions. So that is why I put this question like that.

To what extent does the survey methodology play a crucial role in employees' self-assessment of their behavior, particularly regarding compliance with cybersecurity policies?

John: It's critical in the sense, I think you have to understand, I think, maybe the limitations of the survey methodology. So right in our study and in a number of them, we're asking people to self-report on their compliant behavior with regard to cybersecurity policies.

I think we make the point in the paper and they make this point sometimes in the experience sampling literature that when you're, I think we argue that this is the best or one of the better ways to get at it.

Obviously, the Holy Grail would be if you get actual logs of data on a daily basis and see if people are complying with aspects of cybersecurity policy. But absent that, asking people over the course of the week is generally considered an acceptable way of doing it.

Go ahead, Alex.

Alexander: I did want to interrupt you, John. Just to add one thing, so if you want to, well, for example, test psychological constructs like neutralization, you just have to ask people. You cannot extract this from any locks.

You can extract the compliance behavior maybe, right? But if you want to test theories, psychological theories, and I think there are plenty in the behavioral cybersecurity research, right?

I think there is no other way around, right? You need the survey methodology as one critical component.

John: Yeah, that's a great point. I thought maybe you're getting at with your question, which would be a good one if someone were to say, well, why did you guys use a survey for this?

I think Alex's answer is a good one. For specifically, we're testing the effect of neutralization, which is a psychological construct, and I don't know if there's really an objective measurement of that. So at least for that part of the relationship, we would need a survey to get at that neutralization. I agree. I don't know if anyone's come up with like physiological measures for that.

Kaushik: *Can I simply put like this that for behavioral cybersecurity research, I mean, the primary survey method is one of the standardized thing that we can take into account?*

John: Survey methodology? Yes, it is. I will say the research designs that employ the survey methodology are getting more rigorous. I mean, ours is the experience sampling design is, in some ways, you could argue more rigorous than just the single shot snapshot one time, right?

So multiple data collections, some of these designs are getting pretty sophisticated where you can better assess causality and temporal precedence. So survey methodology in general is still, I think, probably one of the more popular approaches in behavioral cybersecurity research.

I think the designs are getting more robust and more sophisticated, I would argue, but still serve. A lot of times people say, well, I've heard the comment, is survey research dead? Is it sort of outdated? And I think it's more a case of we just have to get more sophisticated with how we use survey research.

I know myself, I like to look at some of the research that's being done in like industrial organizational psychology and in journals like the Journal of Applied Psychology, and they have some really interesting designs where they're still using sort of research methods, or excuse me, survey methods.

Alexander: Or what you can also observe is that the survey methodology is increasingly combined and mixed with other types of methods, right, in mixed or multi-methods research designs.

So I would totally agree with John. I think, well, in our days, you have to, well, make the research design a bit more sophisticated in, well, in a specific dimension.

Either you are conducting multiple measurements over time, like experience sampling, right, or combine it with other methods, be it an experiment or a qualitative study. So I think this is what we are currently seeing in terms of the survey methodology.

I think that's in general in terms of the IS field, not just with regard to behavioral cybersecurity research, but as we pointed out in the paper, I think in one of the appendix articles, there was, there's sort of a history of this snapshot single survey approach in the behavioral cybersecurity literature.

So there's definitely room for more what we would call sophisticated approaches, and research designs with surveys.

Interviewers: Yeah, thanks, Professor.

Mayukh: So we move on to the next section. It is basically reflective questions for the authors.

Can you recall a specific moment during the collaborative writing process when you had differing perspectives on how to interpret a set of data or derive conclusions from the results? So do you, did you have any different perspective, and then you converged?

John: Do you remember anything, Alex, or we were really different?

Yeah, I remember sort of us going through the data and again, we initially had the cybersecurity fatigue longitudinal study in there, which was, that study was conducted over the course of months, whereas we ended up going with the experience sampling study for neutralization theory, which was over a weekly time frame.

But I don't think, I don't think we had vastly different perspectives. We did, between the two of us, sort of do some multiple analysis on the data until we, I think maybe, yeah, we were, one of the things I do remember was we were finding some differences over time in the data.

So something was happening in the data where it was like, when we broke it down and used this idiographic approach, we were finding some differences. I think initially we didn't quite know how to make sense of that and ultimately we ended up putting in a little bit of a theoretical perspective, how well the idea that neutralization theory acts differently or the strength of the theory is different over the course of the week.

And I think we draw on the, we draw on the idea of ego depletion in the paper, which was, we did initially have that in there, but I, to answer your question, I don't think we had really divergent perspectives. It was sort of more iterative as we were, you know, eventually, we came up with that to try to explain what we were finding.

Because initially, I also think when we started this out, we were just trying to make the point that these theories might operate differently over time. If you look at them over time, if you, if you apply an ideographic approach and then it became, okay, now explain why this is happening differently over time.

So that was sort of the iterative process I think we went through if I recall correctly. When I say if I recall, because you have to remember these papers, it's been a couple of years now, right? Since we, once you see them at least, so we sort of have to refresh our memory a little bit.

Mayukh: Anything to add Professor Alex?

Alexander: No, no, I think we had no disagreements or divergences in terms of how we interpreted our data. I just can remember another thing, but not related to the data, but more on how many studies to include in our paper. We already mentioned it. We had the cyber security fatigue on the one side and then neutralization theory on the other side.

I think we had in one of our revisions, I think we had the two studies in the paper, right? And I think, well, we had a discussion and we had a discussion in the team, should we include both, or should we focus on only one study to showcase our key messages in the paper? But I think it wasn't a big issue, right? But we had to struggle with this question. Should it be a multi-study or well, only one science and well, we ended up well, including only the neutralization study in the paper.

John: And if I recall, I think the review team sort of gave us, they didn't really give us, they didn't say either way exactly what to do. They said, just sort of think about this, maybe focus on the one where you can more strongly theoretically justify it. Whereas I think we thought the fatigue study was a bit more exploratory.

It was sort of demonstrating that there might be changes over time, but sort of gets to the decisions that an author team has to make during the course of the review process. Explain why you did what and what your thinking was behind what you did and what changes you made in the paper.

Interviewers: Got it, Professor.

Mayukh: *Looking back, is there a particular aspect of the research article that you now believe could have been approached or explained differently to enhance the clarity or impact of the paper?*

John: We, I'll say one thing, we've been mentioning that there was this sort of this other study, the cybersecurity fatigue study that was cut out. And initially, I liked it because it was an idiographic approach over a longer time frame. So I think our thinking was we'll have one study that's sort of a shorter time frame, weekly approach, and then one that's a longer time frame.

They are both examples of this idiographic approach of within-person processes or changes over time. But again, we had to sort of make some decisions on what to cut from the paper.

So, if we had all the space in the world the review team would have liked it and I think it would have been interesting to have that other example in there of a more longitudinal study than the one we have over a longer time period. But I think we'll find something to do with that additional data.

Alexander: Yeah, I have nothing to add.

I think that would have been great, right? To have two studies with different time periods, right? One with quite a short time period and one with a more extended time period. That would also have helped us to showcase the range of longitudinal studies and how the idiographic approach can be applied to these different time ranges. This would have been great, right?

But yeah, as John already said, we did not have the space in the end. And I think it's a tighter paper, right, to have one study.

John: Yeah, I agree.

Interviewers: Yeah, thanks, Professor.

Mayukh: *Sir, you mentioned like the reviewers were very direct and clarified. Was there a specific instance where external feedback or peer reviews influenced significant changes to the manuscript, leading to new insights or improved clarity in conveying your research findings?*

Alexander: Yeah, I think we have already mentioned this in our answers before there was kind of direct feedback, but not in the sense that they said what we have to do, but they gave us good questions.

For example, think about choosing between one of the two studies that you have included in this round maybe it's better to focus on one study and make every point that you want to make crystal clear with the one study right at hand.

And so, I think we had really good reviewers that guided us in, well, in a particular direction.

John: And editors too, right, good associate editor and a senior editor that were.

That's right. It was pretty clear, not exactly what to do, but it was in terms of the direction to take, they gave us pretty clear guidance and I thought it was excellent.

Mayukh: Thanks, professor.

So, over to you, Kaushik, for the last sections on the guidance for the young scholars, would you like to give?

So, Kaushik, so just we mentioned, we have like six questions in this. We have five here and another one as we have last added. So we can take up three or three from that so that for the, I mean, respect of time of our professors. I'll take because, because many of the questions I think professor have already answered earlier.

Kaushik:

Yeah. I'll make it very specific.

What advice would you provide to young researchers venturing into the field of behavioral cybersecurity research, especially in terms of choosing between a nomothetic and idiographic approach for their studies, particularly for those coming from other fields like marketing or psychology?

John: Well one thing, you know, you mentioned sort of if you're thinking about choosing between nomothetic and ideographic approach. As we talk about in the paper, the ideographic approach is beneficial when you think there's the notion of within-person change over time.

So maybe not all theories or phenomena, what you're studying that plays a role. But if within-person processes change over time or play a role, I think that lends itself to this idiographic approach. We talk about that a little bit or probably more than a little bit, in the last section of the paper, we try to articulate where we think this idiographic approach is perhaps most valuable when time matters.

About your question for scholars coming from marketing or psychology. Just as a general comment, I think it would be important for them to understand the current conversation and the information systems literature with regard to behavioral cybersecurity to sort of know what is happening in the literature, and what are the current approaches.

So if I was someone, you know, from that discipline, and I wanted to perhaps publish an MIS quarterly, I think it's important, again, to understand sort of the current conversation happening in our discipline, not that it has to be written exactly like an IS paper, but at least sort of situating

what you're studying within, you know, what's going on amongst IS scholars who are studying the topic.

That's a more general piece of feedback, but at least on that, that's what I would say with regard to that particular part of your question.

Alexander: Yeah, maybe one point to add to this.

I think younger researchers have to be pretty clear about the effort they have to invest in these methods. We've covered it already, that ideographic approach that they oftentimes entails collecting, well, multiple measurements over time.

So I would say in general, the research designs are some sort of more and more complex, more sophisticated, right? And I think we have to understand and learn a lot if you want to apply, for example, experience sampling methodologies.

So I think one of the recommendations I would give is maybe if you're totally new, right, to, well, the nomothetic and idiographic approach, I would maybe start, you know, in a first project with the nomothetic approach, right?

And well, to get to know how, well, a survey design, for example, looks like, right, for a cross-sectional study. But in the next step, right, you can extend your knowledge into the idiographic field and methods as a next step.

Kaushik: Thank you so much. So my next question is, this is a very direct question related to the survey.

During the survey, you rejected 32 candidates and removed 70 participants. Apart from the reasons stated in the paper, did you observe any specific patterns in the rejected candidate data (e.g., age group, gender, industry of the participant)?

John: I have a question.

The question was, were there any patterns in terms of the rejected candidates based on things like age group, gender, industry, right, so those types of typical demographic characteristics?

Kaushik: Yes.

John: No, I mean, one of the things that should be brought up with doing this type of research using the idiographic approach, in particular experience sampling, is you do get attrition, right?

It's hard to keep that many people, you know, I guess, involved in the study, particularly as they get longer periods of time, particularly, you know, over the course of weeks. And some of these designs of ours, we just had one survey per day, but many experience sampling approaches will have multiple surveys per day as well. So getting that commitment up front from the participants is critical.

For ours, from what I recall, other than the ones that failed attention check questions, or simply didn't answer, didn't continue on in the study, and answer enough surveys, there weren't any systematic differences based on the demographic characteristics. We did look for that.

So it was sort of just certain people, and who knows, maybe it was the time commitment involved or something, just fell out.

I've done, there was another, not for this project, but there was another experience sampling project I was involved in before where I got participants from Mechanical Turk. And I remember it was sort of, it became almost personal where I would, there weren't that many, there were about 50 of them in the whole study, but it was over the course of weeks.

But I would email people reminders, right, since there was that few of participants, I would email them reminders at the end of that day, like, did you get the survey for today?

So there was probably more involvement than a typical study in trying to keep participants involved over the course of a longer period of time. That's one piece, that's one experience I had previously.

Alex, I don't, do you recall anything in the data for this one where we have anything sort of systematic in terms of people that were rejected?

Alexander: Well, no, I think, I do not remember any significant differences, right?

John: Yeah, the biggest thing I would say, for these types of studies, there is some upfront work and sort of almost establishing the relationship. And like in your cover letter and in the description of the project upfront and a consent form explaining to the participants that this is a significant commitment to get the full incentive you have to participate in the full study and here is what it is going to entail.

So I think, for those people that understand sort of the parameters of the study upfront and are willing and accepted that that's who you want to have, and then the key would just be explaining or figuring out that those people are not different from the general population.

So the people that choose to partake in this sort of intensive study are not different than the population that you're trying to get your sample from. You don't want any systematic differences there. Yeah, it's challenging.

And that's, I think, sort of a little bit similar to the previous question, when Alex was giving some advice for young scholars who might want to get involved in research like this. You do your cross-sectional survey and then you expand your knowledge base and start doing studies like this, and at least for me, some of the knowledge gained was just actually through going through the process and conducting studies like this.

And it's almost a little bit of trial and error.

Kaushik: You also raised one pain point of getting the survey data.

Mayukh: So Kaushik, I will take up the last question.

Kaushik: Yeah, please.

Mayukh: *So this is the question we mailed you this morning, sorry for the late interruption, because there are certain events that evolved recently.*

So our last question would be like, we observed two worldwide significant cybersecurity events recently.

One was the Microsoft and CrowdStrike blue screen of death that caused a routine bug and which crippled the entire systems in the world, which runs on recent Windows and Azure.

And there was another event that was very specific to India in this case, because it caused a major cryptocurrency hack of around 230 million from an Indian crypto exchange called WazirX, which was a multi-sig wallet.

So our question is like, I mean, in your paper, you describe three contexts at the end as you describe.

So we wanted to understand like as a young scholars, when we try to approach such black swan events and do post event analysis, how can your paper help us in understanding it from a behavioral research perspective?

Alexander: So maybe I'll just go first.

So as you know, the idiographic approach and, and behavioral cybersecurity research is, is primarily about individual behavior. And the idiographic approach is, then also about a longitudinal within-person view. So how does individual behavior evolve over time, right?

And, maybe also to think about what kind of vulnerabilities are there within persons over time. And if you look at least at the route strike event, you have to ask yourself where in this process are individuals involved?

Or such a buck can have disastrous implications, and consequences. And I think in this case, I don't know whether, whether I'm really deep into the background of this. But I think here the testing behaviors of software developers or testers will be interesting to monitor over time. Whether there are some critical oversights in their work, or if there are biases or something like that.

So how does behavior of individual developers testers change over time, are there any negligence or negligent behaviors, or do they miss something over time, right?

So other specific testing routines employed, but not at any one single point in time. The idiographic approach is more about monitoring, upholding routines that increase the likelihood that specific quality levels are ensured. So I think this would be one thing that how you could as a young researcher look into this to prevent, or at least mitigate the risks of such a black swan events.

John: Yeah, I don't have much to add to that other than time matters and it perhaps is, do people become more negligent or less security conscious, over time. We're on different days of the week, it is interesting that this event happened on a Thursday night or Friday, closer to the end of the workweek, which is we talked about in our paper that people are more susceptible to it.

Alexander: I didn't know that. Yeah, right.

John: Yeah, I was affected by it personally. I woke up in the United States here Friday morning. I went to get on my computer and do my work and I had that blue screen. It caused a few hours of being non-productive, but yes, it was an end-of-the-workweek event.

We don't know exactly what happened. But to Alex's point, there is at least the monitoring aspect of it, perhaps, you know, time of the week maybe played a role potentially in making that happen.

The other one, was the event in India, which was with them, I wasn't familiar with it, but I read about the crypto wallet getting hacked into.

Mayukh: Yes, it was a multi-signature wallet, but still around half of their AUM got wiped out by a crypto hack.

So they had around 530 million dollars and around 230 million dollars were completely wiped out by a hacker. So it's an ongoing event still now.

So they have raised a bounty also to catch the hackers, but it's still an evolving event as of now. So the WazirX is the platform.

John D'Arcy: OK. So try drawing that a little bit into what we are talking about in our paper and the idea of employee security-related behavior and how it's subject to time-based influences. We do know even with external hacking events, oftentimes they come to fruition. The root cause is still human.

We have these hacking events that come from the outside of the organization. However, there might be some type of vulnerability within the IT systems of the organization that is because there was some type of oversight or negligence on the part of the human.

For example, a patch wasn't updated on a critical server. To the extent that at certain points during the week or at certain or different time periods this might be more or less of an issue, perhaps that could play a role in making the organization more or less susceptible to external hacking behavior at different time periods.

I think it's probably you can draw maybe stronger, more direct insights for the CrowdStrike one, given that there was probably some type of lack of monitoring, right?

As Alex mentioned, right, that likely somehow, you know, a software update that was buggy got out, right? And it shouldn't have.

Mayukh: Thanks, Professor, for your amazing insight.

Kaushik: Thank you, Professor.

Alexander: Thank you for this great initiative.

John: Thanks for the work you're doing.

Kaushik: It was a very enriching and enriching discussion.

So it provided a lot of valuable insights into the area of this research, into the area of information systems research. So with your permission, let us call it a day, let us end this session.

John: Thank you.

Alexander: Thank you.